

Product Requirement Document (PRD)

InuaBiz Architecture & Implementation: KRA eTIMS / ETR Integration Module

Project: InuaBiz POS & Business ERP

Author: Lead Systems Architect

Target Region: Kenya (KRA Compliant)

Status: Approved / Ready for Build

1. Executive Summary & Objectives

The Kenya Revenue Authority (KRA) mandates electronic tax invoice generation through the **eTIMS** system. This PRD specifies the technical requirements for embedding a software-driven eTIMS compliance module directly into the **InuaBiz** platform.

This module eliminates the need for expensive physical Electronic Tax Register (ETR) hardware by utilizing direct, automated API communications (OSCU/VSCU) with KRA's Online/Virtual Sales Control Units. Furthermore, it introduces a tiered SaaS monetization model managed seamlessly via the InuaBiz Super Admin portal.

2. Core Features & Functional Scope

Key Architectural Pillars:

- **Hardware-Free ETR Generation:** Direct issuance of compliant electronic tax receipts complete with KRA Control Numbers, QR codes, and digital signatures.
- **Dual Operating Modes (OSCU & VSCU):** Support for real-time cloud API submission (OSCU) as well as resilient local/offline queueing (VSCU).
- **Super Admin Gating & Monetization:** Tiered feature toggling managed via the Super Admin portal, gating eTIMS capabilities based on subscription tier.
- **Automated Compliance Enforcement:** Real-time credential validation, automatic invoice state syncing, and graceful handling of subscription expiration/delinquency.

3. Subscription Tiering & Commercial Model

InuaBiz Pricing Architecture

Subscription Plan	Monthly Fee (KES)	Included Features	eTIMS / ETR Access
Standard Tier	KES 3,000 / month	POS, Inventory Management, Sales Analytics, Multi-branch reporting.	DISABLED / LOCKED
Compliance Tier (eTIMS)	KES 4,500 / month	All Standard features + Direct KRA eTIMS API Integration, Automated ETR Receipting, Audit Logs.	ENABLED / ACTIVE

4. Admin & User Interface Workflows

4.1 Super Admin Dashboard Gating

- The Super Admin can toggle the eTIMS Module setting per tenant/merchant branch.
- When enabled (upon receipt of KES 4,500/mo), the merchant UI unlocks the **KRA Settings** tab and activates automated receipt signing on checkout.

4.2 Tenant / Merchant Settings Interface

- **Locked State (Standard Tier):** The eTIMS settings tab displays a promo card: *"Upgrade to Compliance Tier (KES 4,500/mo) to unlock automated KRA eTIMS integration."* Input fields are disabled.
- **Unlocked State (Compliance Tier):** The tenant inputs their KRA PIN, Branch Code, Serial Number, and API Bearer Keys provided via the KRA Taxpayer Portal.

4.3 Subscription Expiration & Delinquency Logic

- If a tenant fails to renew their KES 4,500 subscription, the system automatically transitions the eTIMS module to a **Suspended / Locked** state.
- Checkout transactions will default to non-fiscal standard receipts with a warning banner: *"KRA eTIMS transmission suspended due to inactive subscription."*

5. Technical & Integration Architecture

The integration connects the InuaBiz POS sales engine to KRA eTIMS via an asynchronous background service layer.

1. Point of Sale

Cashier completes sale on InuaBiz UI. Local invoice payload generated.

2. InuaBiz Service Layer

System checks tenant tier. If active, queues payload to KraIntegrationService.

3. KRA OSCU / VSCU API

Payload sent with Bearer Token & Idempotency Key. KRA signs receipt.

4. Receipt & QR Code

InuaBiz receives QR URL + Fiscal Code, attaching them to customer receipt.

5.1 API Data Payload Example (InuaBiz -> KRA eTIMS)

```
{
  "tin": "A012345678Z",
  "bhfId": "00",
  "invcNo": 10452,
  "orgInvcNo": 0,
  "custTin": "P051234567A",
  "custNm": "Walk-in Customer",
  "salesTyCd": "N",
  "rcptTyCd": "S",
  "pmtTyCd": "01",
  "itemList": [
    {
      "itemSeq": 1,
      "itemCd": "KE1NTX00012",
      "itemNm": "Amoxicillin 500mg",
      "pkgUnitCd": "CT",
      "qty": 2.0,
      "prc": 1500.0,
      "splyAmt": 3000.0,
      "vatTaxblAmt": 2586.21,
      "vatAmt": 413.79,
      "totAmt": 3000.0
    }
  ]
}
```

5.2 Detailed Integration Steps for Engineers

Step	Task Name	Technical Implementation Details
1	Sandbox Registration	Obtain OSCU/VSCU sandbox credentials & API keys from the KRA Taxpayer Sandbox Portal.
2	Backend Module Setup	Create KraService in Node.js/Python backend. Implement AES-256 encryption for stored API tokens.
3	Database Schema Update	Add etims_enabled, kra_pin, branch_code, and device_serial to shops table. Add fiscal fields to invoices table.
4	Validation & Error Handling	Capture API error codes (e.g., invalid PIN, tax mismatch). Return human-readable UI alerts on settings view.
5	Webhook / Status Listener	Implement asynchronous webhook endpoint to handle delayed fiscal signing acknowledgments from KRA.

6. Security, Compliance & Audit Trail

- **Data Encryption:** All KRA secrets and communication keys must be encrypted at rest using AES-256 and in transit via TLS 1.3.
- **Idempotency:** Every invoice submission must carry a unique UUID idempotency key to prevent double fiscalization on connection retries.

- **Audit Logging:** Maintain an immutable log of all request/response payloads exchanged with KRA for a minimum of 7 years for tax audit compliance.